CDPM2 UMAT Manual

Fortran implementation for implicit Abaqus/Standard

2026-10-14

Origins, implementation, representative verification, built-in CDP comparisons and application limits.

English and Japanese editions have matching sections, figures, results and references.

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Contents

1 CDPM2 origins and this project 3
1.1 What this project develops . . . 3
2 Terminology and distinction from CDP 4
3 CDPM2 material model 5
3.1 Stress, strain and damage . . . 5
3.2 Stress invariants and yield condition . . . 5
3.3 Nonassociated flow and dilation . . . 6
3.4 Hardening and pressure-dependent ductility . . . 6
3.5 Evolution of hardening and damage histories . . . 7
3.6 Tensile softening, fracture energy and characteristic length . . . 8
3.7 Compressive softening and unloading/reloading . . . 8
3.8 Inputs, viscosity and scope . . . 8
4 Implicit UMAT update and viscosity 9
4.1 Numerical stability and tangent checks . . . 9
5 Representative implementation verification 10
5.1 Verification of the numerical update and restart extension . . . 11
6 Triaxial compression at different confinement 12
7 High-pressure hydrostatic compression 13
8 Three-point bending of a notched beam 14
9 Influence of viscosity and loading time 15
10 Installation and execution 16
10.1 Build and minimal examples . . . 16
10.2 Mesh geometry from the results file . . . 16
10.3 Restart . . . 17
10.4 Contact stiffness multipliers . . . 17
11 Material inputs and state variables 18

11.1 Material constants 18
11.2 State variables 19
11.3 Numerical acceptance controls 19

12 Application limits and distribution 20
12.1 Tangent and energy limitations 20
12.2 Contribution and distribution 20

1 CDPM2 origins and this project

CDPM2 is a concrete damage-plasticity constitutive model published in 2013 by Peter Grassl, Dimitrios Xenos, Ulrika Nyström, Rasmus Rempling and Kent Gylltoft[1]. It extends the 2006 damage-plasticity model of Grassl and Jirásek, called CDPM1 in the 2013 paper. CDPM2 separates tensile and compressive damage and extends the relationship between hardening and damage[1, 2].

Plasticity represents irreversible deformation; damage represents stiffness degradation. Their combination describes tension, compression, confinement effects, unloading and reloading.

1.1 What this project develops

This project implements the existing CDPM2 model in Fortran as an implicit UMAT for Abaqus/Standard. It does not propose a new CDPM2 constitutive model. The public OOFEM implementation is the reference for the port. This project supplies the Abaqus stress, strain, state-variable, rotation and tangent interface[3]. It is not an official OOFEM or SIMULIA UMAT.

- CDPM2 paper: the constitutive model defining the material response.
- OOFEM: the existing public reference implementation.
- This UMAT: the Fortran implementation for the implicit Abaqus/Standard solver. Viscous relaxation is an added extension.

The objective is a traceable constitutive implementation with practical convergence. No parameter fitting to experiments was performed for these comparisons.

2 Terminology and distinction from CDP

CDP Concrete Damaged Plasticity. In this report, CDP means the built-in Abaqus concrete model[9].

CDPM2 Concrete Damage Plasticity Model 2, the separate model of Grassl and coauthors. It is not version 2 of Abaqus CDP.

Plasticity Irreversible deformation remaining after unloading.

Damage An internal variable describing stiffness degradation; 0 denotes undamaged material and 1 complete damage.

Hardening and softening Increasing and decreasing resistance as deformation progresses.

Dilation Plastic volume expansion. The CDP dilation angle and the CDPM2 flow parameter are not directly interchangeable.

Confinement Compressive pressure applied laterally to the specimen.

Viscosity In this UMAT, an added time relaxation of plastic strain and damage. It can aid convergence but also changes the response.

UMAT and DDSDDE The user-material subroutine and the material tangent matrix supplied to the global Newton iteration.

Both models combine plasticity and damage, but their yield surfaces, hardening, flow rules and softening-damage relationships differ. CDPM2 has pressure-dependent plastic flow. Equal parameter or viscosity numbers do not imply equal responses.

3 CDPM2 material model

3.1 Stress, strain and damage

CDPM2 combines irreversible plastic deformation with damage-induced stiffness loss. The equations below follow Grassl et al., Sections 2–3 (preprint pp.3–17), and the adopted OOFEM implementation[1, 3]. Implementation extensions are identified explicitly. The topics follow the Abaqus CDP material description[9]; its constitutive equations are not used for CDPM2.

Tension is positive; $f_t, f_c > 0$ are positive strength magnitudes. For isotropic elasticity with Young's modulus E and Poisson's ratio ν,

$$\bar{\boldsymbol{\sigma}} = \mathbb{C}_e : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p), \tag{1}$$

$$\boldsymbol{\sigma} = (1 - d_t)\bar{\boldsymbol{\sigma}}^+ + (1 - d_c)\bar{\boldsymbol{\sigma}}^-. \tag{2}$$

$\bar{\boldsymbol{\sigma}}$ is effective undamaged stress, whereas $\boldsymbol{\sigma}$ is returned to Abaqus. The spectral splits replace each principal stress by $\max(\bar{\sigma}_i, 0)$ or $\min(\bar{\sigma}_i, 0)$ and transform back to the tensor. $d_t, d_c = 0$ denote undamaged material and 1 complete damage. Damage is not plastic strain or measured crack opening.

3.2 Stress invariants and yield condition

Mean effective stress s, deviator $\bar{\boldsymbol{s}}$, deviatoric norm ρ and Lode angle θ are defined by

$$s = \tfrac{1}{3}\operatorname{tr}\bar{\boldsymbol{\sigma}}, \qquad \bar{\boldsymbol{s}} = \bar{\boldsymbol{\sigma}} - s\boldsymbol{I}, \tag{3}$$

$$J_2 = \tfrac{1}{2}\bar{\boldsymbol{s}} : \bar{\boldsymbol{s}}, \qquad J_3 = \det \bar{\boldsymbol{s}}, \tag{4}$$

$$\rho = \sqrt{2J_2}, \qquad \theta = \tfrac{1}{3}\arccos\left(\frac{3\sqrt{3}J_3}{2J_2^{3/2}}\right). \tag{5}$$

Pressure is $p = -s$; confinement makes s more negative. $0 \le \theta \le \pi/3$ on regular returns. Hydrostatic vertices ($\rho = 0$) use a dedicated return, not division by zero. With hardening functions $q_1(\kappa_p), q_2(\kappa_p)$,

$$B = \frac{s}{f_c} + \frac{\rho}{\sqrt{6}f_c}, \qquad A = (1 - q_1)B^2 + \sqrt{\tfrac{3}{2}}\frac{\rho}{f_c}, \tag{6}$$

$$F = A^2 + m_0 q_1^2 q_2 \left(\frac{s}{f_c} + \frac{\rho r(\cos\theta)}{\sqrt{6}f_c}\right) - q_1^2 q_2^2, \tag{7}$$

$$m_0 = \frac{3(f_c^2 - f_t^2)}{f_c f_t}\frac{e}{e+1}, \tag{8}$$

$$r(c) = \frac{4(1-e^2)c^2 + (2e-1)^2}{2(1-e^2)c + (2e-1)\sqrt{4(1-e^2)c^2 + 5e^2 - 4e}}. \tag{9}$$

e controls the deviatoric section; m_0 is calculated from strengths and e, not an input friction angle. Admissibility is $F \le 0$, $\dot{\lambda} \ge 0$, $\dot{\lambda}F = 0$. At $q_1 = q_2 = 1$, the surface becomes the Menetrey–Willam strength envelope. Confinement dependence is contained in the surface, not imposed by changing f_c.

3.3 Nonassociated flow and dilation

Plastic strain follows a potential g different from F:

$$\dot{\boldsymbol{\varepsilon}}^p = \dot{\lambda}\,\frac{\partial g}{\partial \bar{\boldsymbol{\sigma}}}, \qquad \frac{\partial g}{\partial \bar{\boldsymbol{\sigma}}} = \frac{g_s}{3}\boldsymbol{I} + g_\rho \frac{\bar{\boldsymbol{s}}}{\rho}. \tag{10}$$

A potential consistent with the distributed regular-branch gradient is

$$g = A^2 + q_1^2\left[\frac{m_0\rho}{\sqrt{6}f_c} + A_g B_g \exp\!\left(\frac{s-s_0}{B_g f_c}\right)\right], \tag{11}$$

$$s_0 = f_t q_1 q_2/3, \qquad A_g = 3f_t q_1 q_2/f_c + m_0 q_1^2/2, \tag{12}$$

$$B_g = \frac{q_1 q_2(1+f_t/f_c)}{3[\ln A_g + \ln(D_f+1) - \ln(2D_f-1) - \ln(3q_1q_2 + m_0 q_1^2/2)]}. \tag{13}$$

For $M = A_g \exp[(s-s_0)/(B_g f_c)]$, the gradients used in the code are

$$g_s = 4(1-q_1)AB/f_c + q_1^2 M/f_c, \tag{14}$$

$$g_\rho = \{A[4(1-q_1)B+6] + m_0 q_1^2\}/(\sqrt{6}f_c). \tag{15}$$

Unlike F, g does not depend on Lode angle. The q_1 factors follow the adopted OOFEM `computeDGDInv` generalization; at $q_1 = 1$ they reduce to the 2013 paper's Eqs.(22)–(29). The pre-peak generalization is not presented as a verbatim equation from the paper.

$D_f = -\dot{\varepsilon}^p_{\text{lateral}}/\dot{\varepsilon}^p_{\text{axial}}$ is the dimensionless ratio in uniaxial compression for $q_1 = 1$, not a dilation angle in degrees. At fixed hardening state, increasing confinement reduces M and the volumetric-flow contribution: high confinement suppresses dilation. The lateral/axial ratio is not constant on arbitrary multiaxial paths. Similarly, a constant CDP dilation-angle input does not imply a constant plastic strain ratio in every CDP stress state. The calibration gives zero lateral plastic strain in uniaxial tension for $q_1 = 1$; this does not mean zero total lateral strain. Nonassociated flow generally gives an unsymmetric tangent; the examples request `UNSYMM`.

3.4 Hardening and pressure-dependent ductility

The accumulated variable κ_p controls surface shape and size:

$$q_1 = \begin{cases} q_{h0} + (1-q_{h0})(\kappa_p^3 - 3\kappa_p^2 + 3\kappa_p) - H_p(\kappa_p^3 - 3\kappa_p^2 + 2\kappa_p), & 0 \le \kappa_p < 1, \\ 1, & \kappa_p \ge 1, \end{cases} \tag{16}$$

$$q_2 = \begin{cases} 1, & \kappa_p < 1, \\ 1 + H_p(\kappa_p - 1), & \kappa_p \ge 1. \end{cases} \tag{17}$$

q_{h0} defines the initial surface and H_p continued effective-stress hardening. Nominal stress may soften while q_2 grows, because damage increases; plastic hardening is not stopped by nominal softening.

3.5 Evolution of hardening and damage histories

On regular returns,

$$\dot{\kappa}_p = \frac{\|\dot{\boldsymbol{\varepsilon}}^p\|}{x_h(s)}(2\cos\theta)^2, \qquad R_h = -s/f_c - 1/3, \tag{18}$$

$$x_h = \begin{cases} A_h - (A_h - B_h)\exp(-R_h/C_h), & R_h \geq 0, \\ E_h \exp(R_h/F_h) + D_h, & R_h < 0, \end{cases} \tag{19}$$

$$E_h = B_h - D_h, \qquad F_h = (B_h - D_h)C_h/(A_h - B_h). \tag{20}$$

A_h, B_h, C_h, D_h control plastic ductility, not damage. The norm is the tensor Frobenius norm: $\|\boldsymbol{\varepsilon}^p\|^2 = \sum_i (\varepsilon^p_{ii})^2 + [(\gamma^p_{12})^2 + (\gamma^p_{13})^2 + (\gamma^p_{23})^2]/2$ in engineering-shear storage. Hydrostatic returns use the reference vertex convention; the regular formula alone does not prescribe the $\rho = 0$ branch.

Let $\varepsilon_0 = f_t/E$ and $Y = s/f_c + \rho r(\cos\theta)/(\sqrt{6}f_c)$. Equivalent strain is

$$\widetilde{\varepsilon} = \varepsilon_0 \left[\frac{m_0 Y}{2} + \sqrt{\frac{m_0^2 Y^2}{4} + \frac{3\rho^2}{2f_c^2}} \right]. \tag{21}$$

It equals $\bar{\sigma}_t/E$ in uniaxial tension and $f_t|\bar{\sigma}_c|/(Ef_c)$ in uniaxial compression. Both reach ε_0 at their reference strengths. This normalization does not replace compressive strength by f_t. For nonzero effective stress, define

$$\alpha_c = \frac{\sum_i \min(\bar{\sigma}_i, 0)^2}{\sum_i \bar{\sigma}_i^2}. \tag{22}$$

Pure tension gives 0 and pure compression 1. Accumulated measures evolve as $\dot{z}_t = \dot{\widetilde{\varepsilon}}$ and $\dot{z}_c = \alpha_c \dot{\widetilde{\varepsilon}}$; their maximum histories are κ_t, κ_c. Damage starts at the corresponding history threshold ε_0. The increment implementation includes reversal handling and a small reference activation tolerance; damage is irreversible.

Two auxiliary histories per mode combine plastic and damaged deformation. On active history branches,

$$\dot{\kappa}_{t1} = \|\dot{\boldsymbol{\varepsilon}}^p\|/x_s, \qquad \dot{\kappa}_{t2} = \dot{\kappa}_t/x_s, \tag{23}$$

$$\dot{\kappa}_{c1} = \alpha_c \beta_c \|\dot{\boldsymbol{\varepsilon}}^p\|/x_s, \qquad \dot{\kappa}_{c2} = \dot{\kappa}_c/x_s, \tag{24}$$

$$\beta_c = \frac{f_t q_2 \sqrt{2/3}}{\rho\sqrt{1 + 2D_f^2}}, \qquad x_s = 1 + (A_s - 1)R_s^{B_s}, \tag{25}$$

$$R_s = \begin{cases} -\sqrt{6}s/\rho, & s < 0, \\ 0, & s \geq 0. \end{cases} \tag{26}$$

The plastic contribution starts after activation; crossing increments use only the post-activation portion. These histories are distinct from κ_p. A_s controls compressive softening ductility. $B_s = 1$ recovers the paper's Eq.(56); B_s is an OOFEM extension. Near $\rho = 0$ the code uses its reference small-denominator convention, not these expressions at zero. In uniaxial interpretation, inelastic strain is $\varepsilon^i = \varepsilon^p + d(\varepsilon - \varepsilon^p)$, not plastic strain alone.

3.6 Tensile softening, fracture energy and characteristic length

A bilinear traction–inelastic-opening law is specified by f_t, knee stress f_{t1}, knee opening w_{f1} and final opening w_f:

$$T(w) = \begin{cases} f_t - (f_t - f_{t1})w/w_{f1}, & 0 \le w \le w_{f1}, \\ f_{t1}(w_f - w)/(w_f - w_{f1}), & w_{f1} < w \le w_f, \\ 0, & w > w_f. \end{cases} \tag{27}$$

Tensile damage is the admissible intersection

$$(1 - d_t)E\kappa_t = T(l_e[\kappa_{t1} + d_t\kappa_{t2}]), \tag{28}$$

$$G_f = \int_0^{w_f} T(w)\, dw = \tfrac{1}{2}(f_t w_{f1} + f_{t1} w_f). \tag{29}$$

l_e converts inelastic strain to equivalent opening. Automatic selection projects initial node coordinates along the maximum-principal-strain direction $\boldsymbol{n}$ at activation: $l_e = \max_a(\boldsymbol{x}_a \cdot \boldsymbol{n}) - \min_a(\boldsymbol{x}_a \cdot \boldsymbol{n})$. The selected length is fixed thereafter. This smeared crack-band regularization reduces tensile mesh-size sensitivity; it is not discrete crack creation or a guarantee of mesh-independent structural results. The code consistently applies the reference activation tolerance to initial traction; the equation above shows the intended softening envelope.

3.7 Compressive softening and unloading/reloading

Compressive damage uses an exponential law in normalized equivalent-strain space, not the tensile opening law:

$$(1 - d_c)E\kappa_c = f_t \exp[-(\kappa_{c1} + d_c\kappa_{c2})/\varepsilon_{fc}]. \tag{30}$$

ε_{fc} is a dimensionless softening scale. The f_t factor is due to normalization. On an active monotonic uniaxial compression branch, the actual compressive stress magnitude is f_c/f_t times this normalized damaged-stress measure, not a compressive strength of f_t. x_s adjusts history growth under confinement. Compression is not regularized by l_e and may retain localization-related mesh sensitivity.

Damage does not heal. For fixed plastic strain and damage, uniaxial unloading slopes are $(1 - d_t)E$ in tension and $(1 - d_c)E$ in compression. Plastic strain shifts the zero-stress intercept. Reversing from tension to compression uses d_c; if $d_c = 0$, compressive stiffness is recovered despite $d_t > 0$. There are no separate CDP-style stiffness-recovery coefficients. Reloading inside the yield domain need not create plastic strain. Damage fields visualize distributed degradation, not discrete crack paths.

3.8 Inputs, viscosity and scope

Use consistent stress, length and time units: E, f_t, f_c, f_{t1} are stresses; w_{f1}, w_f, l_e are lengths; G_f is stress times length. Strains and hardening/dilation/ductility parameters are dimensionless. Optional relaxation time μ is this UMAT's extension, not part of the rate-independent CDPM2 equations; it relaxes plastic strain and damage. It is neither a calibrated strain-rate strength law nor identical to CDP viscosity. The distributed formulation is 3D and isotropic, for the documented Standard cases. Temperature dependence, rate-strength enhancement, rebar bond laws and automatic element deletion are not included.

4 Implicit UMAT update and viscosity

The material equations are described in Section 3. When viscosity is enabled, the final stress uses relaxed plastic strain and damage.

1. Preserve the committed state and check inputs and rotation.
2. Solve the increment endpoint using an elastic predictor and local Newton return mapping.
3. Update damage histories from the plastic history.
4. Apply optional viscous relaxation and calculate stress and DDSDDE.
5. Commit only accepted updates. On failure, retain old stress and state and request an external increment reduction.

The method does not force substepping of every increment. The tangent combines analytical sensitivities with numerical differences at difficult local branches.

$$z^{v}_{n+1} = \frac{\mu z^{v}_{n} + \Delta t\, z^{\mathrm{target}}_{n+1}}{\mu + \Delta t}.$$

μ has the analysis time unit; $\mu = 0$ recovers the non-viscous update. Different relaxed variables mean this extension is not identical to Abaqus CDP viscosity. No artificial damage cap at 0.99, hardening freeze or forced residual-strength plateau is added.

4.1 Numerical stability and tangent checks

The local 4-by-4 linear system is equilibrated by columns and rows before partial pivoting. A componentwise backward-error check precedes unscaling. The same equations are solved; genuinely singular or nonfinite systems remain rejected. No constitutive equation or material parameter is altered. For numerical checks of DDSDDE, compare against the Abaqus tangent definition, not a plain difference of Cauchy stress. In a fixed corotated material-point check this includes the derivative of Kirchhoff stress divided by the volume ratio. Perturbations crossing an activation branch need separate interpretation; a branch crossing is not a smooth derivative test.

5 Representative implementation verification

Check	Previously established baseline results
Material histories	16 histories, 1280 updates. Maximum relative difference in 128 tangent checks: 1.75×10^{-7}.
Rotation	Maximum relative difference 3.93×10^{-15} for the tested rotations and histories.
Rejected-update protection	3282 checks. Rejections are not successful updates.
Difficult saved inputs	1094 inputs: 372 accepted, 717 cutback requests, 5 invalid-input rejections.
Tangents of accepted inputs	370 of 372 completed; maximum relative difference 5.927×10^{-4}. Two perturbation updates were rejected.
Small finite-element models	Seven completed tests including C3D8, C3D6, rotated layouts and two threads.

Test	Increments	Iterations	Rejected attempts
Uniaxial mixed	41	46	1
Shear reversal	40	43	0
Strong displacement history	48	57	8
Same history, half increment	81	88	1
Large initial increment	30	39	19
Rotated C3D6, two elements	40	44	0
C3D8, two threads	40	43	0

All seven tests completed. Rejected attempts precede cutback; they are not failed analyses.

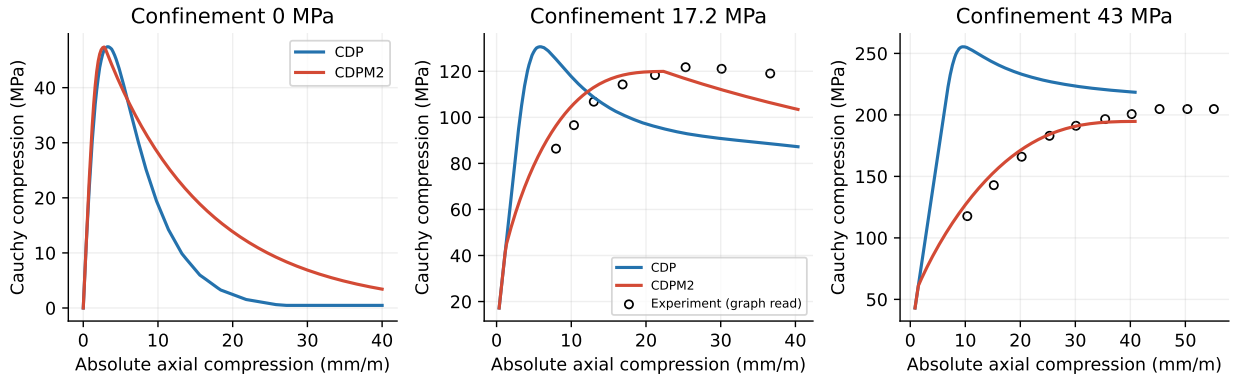
Histories cover tension, compression, hydrostatic and biaxial loading, mixed shear, unloading/reloading and rotations. All 717 cutback inputs could be updated after increment halving from the same fixed old state. This diagnostic does not demonstrate completion of an entire continuous FE history.

Constitutive implementation checks, solver completion and experimental reproduction are separate evaluations. Convergence for arbitrary loading, parameters or meshes is not guaranteed. Failures and known tangent approximations are application limits.

5.1 Verification of the numerical update and restart extension

Saved material-point trials were replayed from the same committed state. Of 167 records, 151 updates were accepted by the updated numerical solver; the previous public source accepted 3. All previously accepted records remained accepted, and their maximum scaled difference was 2.66×10^{-10}. The remaining rejected trials retained their incoming stress and material state. Replaying a rejected trial is not a completed structural analysis. The additional geometry state was also checked in 100 UMAT updates: the original 62 material/work variables, stress and tangent matched the legacy layout exactly. Clearing the geometry provider and reinitializing for restart preserved the state; a rejected large trial retained all 89 state variables. Six Abaqus element comparisons (C3D8, C3D8I and C3D6, each with mu=0 and 0.001) completed. All saved stress, strain and original 62 state variables were compared; the maximum scaled difference from the previous public source was 2.35×10^{-11}. These are small-model regressions, not a general guarantee of convergence. Restart was checked for C3D8, C3D8I and C3D6 with mu=0.001. The normal restart files were moved to another directory without the original FIL; a renamed job resumed the first step and explicitly continued the subsequent steps. All 117 saved frames of stress, strain and all 89 state variables matched the uninterrupted analysis within 5.99×10^{-12} scaled difference. C3D8 and C3D6 were bitwise equal in this comparison. The test covers these static histories, not every procedure or legacy 62-variable restart. The updated 70-, 280- and 1120-element CDPM2 beam examples also completed. Their peak loads were 0.7870, 0.7388 and 0.7221 kN, respectively, consistent with the rounded values reported in Section 8. A two-thread C3D8 test with mu=0.001 matched the serial run exactly in all 44 saved frames, including all 89 state variables.

6 Triaxial compression at different confinement



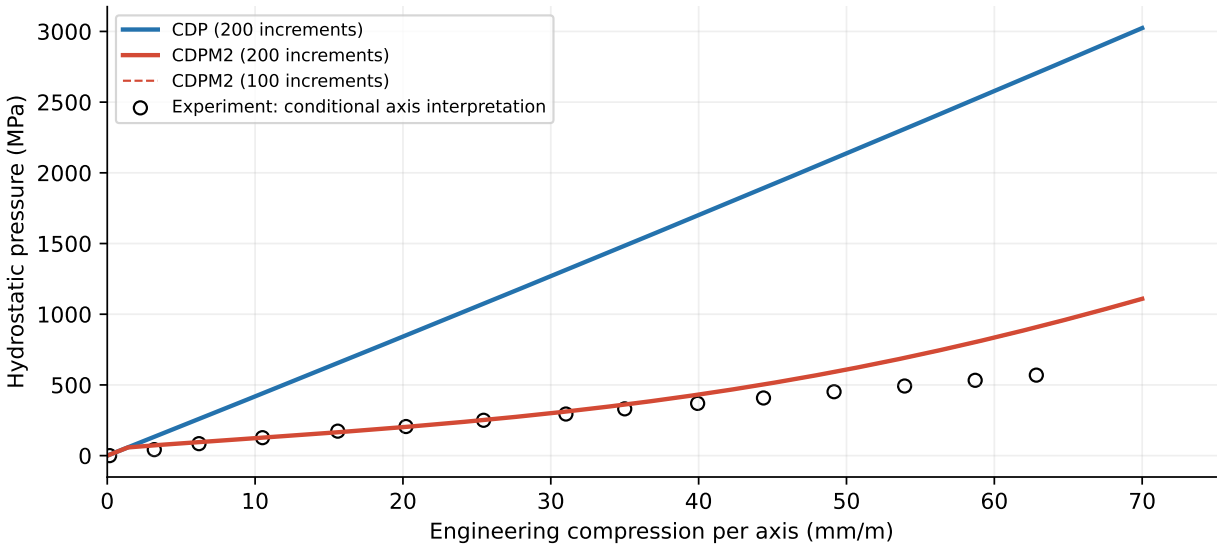
Confinement 0, 17.2 and 43 MPa. Experiment: Imran and Pantazopoulou (1996). Direct reading source: Grassl et al. (2013), Fig.11, PDF p.22[4, 1]. Open markers are approximate plot readings, not original measured data.

100 mm cube, C3D8, axial compression after preloading, NLGEOM=YES, no viscosity. Shared inputs: $E = 30000$ MPa, $\nu = 0.2$, $f_c = 47.4$ MPa, $f_t = 4.74$ MPa. CDPM2: $H_p = 0.01$, $A_s = 15$, $\varepsilon_{fc} = 10^{-4}$, $D_f = 0.85$. CDP: dilation angle 30 degrees, eccentricity 0.1, $K_c = 2/3$.

Confinement(MPa)	CDP max. stress(MPa)	CDPM2 max. stress(MPa)
0	47.423	47.385
17.2	130.648	119.964
43	255.394	194.756

All six analyses completed. The CDPM2 maximum at 43 MPa is near the end of the tested range, not an established ultimate strength. The CDP compression table was scaled for comparison, not fitted to the Imran experiment. Simulation axes are absolute engineering strain and Cauchy stress. Experimental stress and preload strain-zero conventions are unresolved, so the overlay is conditional. The corrigendum cites Imran (1994); correspondence with original data in the 1996 paper remains unresolved. Reading widths are about 0.72 mm/m and 2.04 MPa, not statistical confidence intervals.

7 High-pressure hydrostatic compression

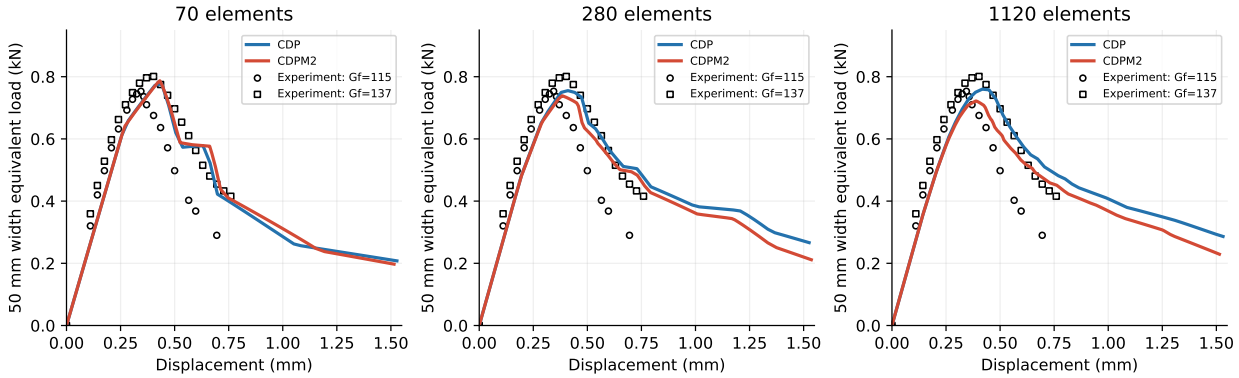


Direct reading source: Grassl et al. (2013), Fig.12, PDF p.22. Experimental reference cited by that paper: Caner and Bažant (2000)[1, 5]. Approximate plot readings; experimental strain interpretation is conditional.

Shared inputs: $E = 25000$ MPa, $\nu = 0.2$, $f_c = 45.7$ MPa, $f_t = 4.57$ MPa, no viscosity. CDPM2 uses $H_p = 0.01$. At 7% engineering compression per axis over 200 increments, terminal pressures are about 3023.78 MPa for CDP and 1109.12 MPa for CDPM2. Damage remains zero in both tests. The CDPM2 terminal difference between 100 and 200 increments is about 0.315%. Independent material-point and FE stresses differ by at most about 2.19×10^{-5} MPa.

The experimental strain axis is conditionally interpreted as engineering compression per axis; the measurement definition is unresolved. Differences remain at higher strains. The adopted OOFEM and original-paper hydrostatic-apex hardening conventions also differ. Exact reproduction of all original curves is not claimed, and inputs were not altered to fit the experiment.

8 Three-point bending of a notched beam



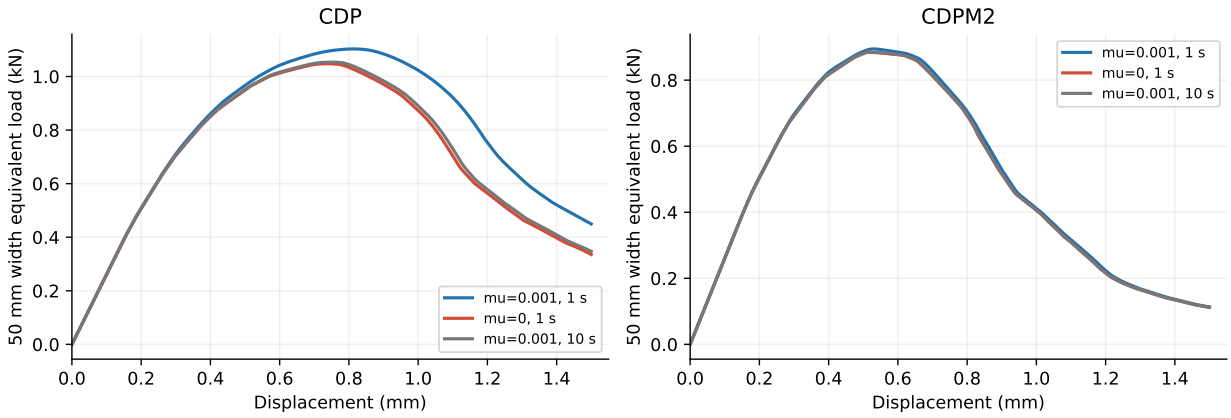
Bilinear-softening paired comparison based on the official 3D inputs: C3D8R, Abaqus/Standard, Riks method. Experiment: Petersson (1981). Direct reading source for experimental markers: Abaqus 2026 Benchmarks, Notched unreinforced concrete beam under 3-point bending, Fig.11[6, 7]. Approximate plot readings; loads scaled to a 50 mm width.

The official 70-, 280- and 1120-element 3D geometry, meshes, boundary conditions and Riks controls are retained. Only the CDP tensile-softening and associated damage tables are changed to the official 2D TS-2 (bilinear) tables. Each CDPM2 case replaces the material in its paired CDP case with the UMAT. **All six cases completed.** This is not a reproduction of the linear-softening results in official Fig.5. Shared inputs: $E = 30000$ MPa, $\nu = 0.2$, $f_c = 30$ MPa, $f_t = 3.33$ MPa, no viscosity, NLGEOM=YES. The official model thicknesses of 20, 10 and 5 mm are retained; loads are scaled to the experimental 50 mm width. The stopping criterion is 1.5 mm loading-point displacement; final accepted displacements are about 1.514–1.536 mm. Riks step time is not physical loading time.

Elements	CDP peak(kN)	CDPM2 peak(kN)
70	0.7773	0.7870
280	0.7552	0.7388
1120	0.7599	0.7221

The TS-2 stress table is 3.33, 0.833 and 0.0333 MPa at openings of 0, 0.025 and 0.198 mm. The CDPM2 bilinear area is matched to its finite-interval area of 0.12697 N/mm, not to the full CDP 1% tail. CDP retains the official 38-degree dilation angle. The UMAT-required hourglass stiffness is 62.5 MPa, based on the same initial shear modulus. The approximate experimental peak is 0.75–0.80 kN. $G_f = 115, 137$ N/m label experimental series, not these simulation inputs. Reading widths are about 0.0195 mm and 0.0130 kN. CDPM2 includes cutback attempts. Its peak and post-peak loads decrease with mesh refinement. No universal model ranking follows from this comparison.

9 Influence of viscosity and loading time



The earlier 280-element C3D8 displacement-controlled beam, a separate configuration from the C3D8R/Riks/bilinear comparison in Section 8. Preserved simulation results only; no experimental markers appear in this figure.

Model	$\mu = 0$,1 s	$\mu = 0.001$,1 s	$\mu = 0.001$,10 s
CDP peak(kN)	1.0475	1.1030	1.0537
CDPM2 peak(kN)	0.8844	0.8946	0.8855

All six cases completed in 200 increments with matched displacement increments. With the same viscosity number, peak increases are about 5.03% for CDP and 1.14% for CDPM2. A tenfold loading-time extension moves results toward the non-viscous response. Viscosity assists convergence but affects the response. Record loading time and increment sizes as well as viscosity. These sensitivities concern one beam configuration and do not establish universal percentage changes.

10 Installation and execution

Scope: Abaqus/Standard 2026, 3D static, NLGEOM=YES, non-perturbation, NDI=3, NSHR=3, NTENS=6, DEPVAR=89, 24 constants, UNSYMM and automatic increments. Intel Fortran tests include C3D8/C3D6, serial and two threads; Section 8 also tests C3D8R and Riks. C3D8I was also checked in small shear and unloading tests. Abaqus handles its internal deformation modes; this UMAT obtains element geometry from the eight corner nodes. MPI, shells, plane stress, axisymmetry, hybrid elements, Explicit, dynamics, thermal coupling and material ORIENTATION remain unverified.

10.1 Build and minimal examples

Supply licensed Abaqus and a compatible Fortran compiler. Download or clone the repository, create a new build folder in its root, copy src/cdpm2.f90 into it and run:

```
abaqus make library=cdpm2.f90
```

In examples, abaqus_v6.env refers to the adjacent build. Initial geometry is read from ASCII FIL output; no geometry table or preparation tool is required. Run:

```
abaqus job=shear cpus=1 mp_mode=threads interactive output_precision=full
```

For the six-node wedge example, use job=wedge in the same folder. Both examples are independent two-element mixed-boundary tests. The six paired notched-beam inputs and extraction/plotting scripts are in examples/notched_beam; its English/Japanese READMEs give the commands and sources. The supplied meshes use explicit data; geometry, connectivity and physical analysis conditions are preserved.

10.2 Mesh geometry from the results file

At the start of the first increment, UEXTERNALDB reads initial global node coordinates (record 1901) and element connectivity (record 1900) from jobname.fil in the output directory. Abaqus resolves mesh generators, includes, node coordinate systems and instance placement; the UMAT no longer parses the input language. The initial geometry is retained in memory and the selected characteristic length is fixed at damage onset.

For your own model, add the settings below to the input file. Place `*FILE FORMAT, ASCII` before the first `*STEP`, outside the descriptive text of `*HEADING`. Place `*NODE FILE` and `*EL FILE` inside every step, without restricting them to a node or element set. The supplied examples already contain these settings. A large output frequency avoids writing results at every increment while retaining the initial mesh information needed before the first material calculation.

```
*FILE FORMAT, ASCII
** Inside each *STEP:
*NODE FILE, FREQUENCY=999999
U
*EL FILE, FREQUENCY=999999
S
```

Supported element types for mesh reading are C3D8, C3D8R, C3D8I and C3D6. The settings above make Abaqus write a text-readable ASCII `.fil` file. The subroutine reads node coordinates and element connectivity from that file; binary `.fil` files are not supported. If the required information cannot be read, an error is written to the log and the material calculation is not performed. Missing geometry is never guessed.

Normally, `jobname.fil` in the current job's output directory is selected automatically, so no file path is needed. Only for a different location, set the environment variable CDPM2_FIL_FILE to an absolute path. With DEPVAR=89, initial geometry is saved in STATEV63–89 and restored by Abaqus restart. No original FIL or external geometry table is required at restart. Legacy DEPVAR=62 still runs, but its restart requires the original explicit geometry table; an old 62-variable restart cannot be converted by changing DEPVAR in a restart deck. Material ORIENTATION and general MPI remain unverified.

FIL regression tests (brick, wedge and two-thread brick) matched the previous input-reader results in all 44 frames

for stress, strain and all 62 state variables. The three FIL beam cases (70/280/1120 elements) also completed; the constitutive update is unchanged. Format references: Abaqus 2026 Output, Records Written for Any File Output Request / About Output; Keywords, *FILE FORMAT.

10.3 Restart

Run the 89-variable shear example, which writes restart data every five increments. From the same folder:

```
abaqus job=shear cpus=1 mp_mode=threads interactive
abaqus job=restart_shear oldjob=shear cpus=1 mp_mode=threads interactive
```

The second command uses examples/restart_shear.inp and the same built UMAT. This restart input includes the later steps explicitly: RESTART READ alone resumes the saved step, not every subsequent step of the original input. Follow Abaqus requirements when transferring the normal restart files to a different folder. Added reference geometry is part of the saved STATEV, so a changed job name or directory does not require a geometry-table tool. An explicit *RESTART, WRITE request is still required. Static restart has been tested; automatic restart for Riks is not claimed.

10.4 Contact stiffness multipliers

*CONTACT CONTROLS, STIFFNESS SCALE FACTOR is an additional multiplier on the scale in *SURFACE BEHAVIOR. For example, 0.1 in each location gives a combined factor of 0.01, not 0.1. Apply a requested factor once, or state the product explicitly. Linear and nonlinear penalty laws with the same factor are not equivalent. Check physical overclosure (negative COPEN), not only the iteration contact-error message. No pullout contact factor is recommended by this manual.

11 Material inputs and state variables

24 material inputs and 89 state variables (62 material/work variables plus 27 immutable geometry values). The example is not a recommended calibration for arbitrary concrete. Example units: N, mm, s; stress MPa. Tension is positive, compression negative; strengths are positive magnitudes. Component order is 11,22,33,12,13,23. Strain/plastic-strain shear is engineering shear; stress shear is tensor shear. Engineering plastic shear squares have weight 0.5 in the norm.

11.1 Material constants

PROPS	Meaning	Example	Unit
1	Young modulus	35892.339	MPa
2	Poisson ratio	0.2	1
3	Compression strength (positive magnitude)	35.892339	MPa
4	Tension strength (positive magnitude)	2.71644205	MPa
5	Yield-surface eccentricity	0.525	1
6	Initial hardening coefficient	0.3	1
7	Hardening coefficient	0.5	1
8	Non-associated flow dilation parameter (not an angle)	0.85	1
9	Plastic ductility ah	0.08	1
10	Plastic ductility bh	0.003	1
11	Plastic ductility ch	2	1
12	Plastic ductility dh	1e-6	1
13	Input characteristic length (distinct from selected length)	20	mm
14	Tensile softening breakpoint stress	0.814932615	MPa
15	Tensile softening breakpoint opening	0.021974493916716565	mm
16	Final tensile softening opening	0.14649662611144376	mm
17	Compressive softening strain scale	1e-4	strain
18	Damage ductility coefficient	15	1
19	Damage ductility exponent	1	1
20	Damage-onset tolerance (not global FE tolerance)	1e-5	1
21	Legacy internal strain cap (compatibility check only)	1e-6	strain
22	Local difference strain scale	1e-9	strain
23	Legacy trial budget (compatibility check only)	20000	count
24	Viscous relaxation time	1e-3	s

PROPS23 does not force 20000 subdivisions. Tensile softening is bilinear traction-opening; compression uses damage history and efc with exponential softening. These are not direct uniaxial total-strain curve inputs. No artificial 0.99 damage cap, hardening freeze or forced residual plateau is added. Distinguish constitutive parameters, numerical controls and user-selected viscosity.

11.2 State variables

STATEV	Meaning
1	State schema identifier 2
2-7	Total strain, engineering shear
8-13	Non-viscous plastic strain, engineering shear
14	Plastic hardening kappa
15-20	Non-viscous effective stress
21-26	Non-viscous damaged stress, distinct from final STRESS
27	Equivalent strain
28	Compression weight alpha
29-30	Accumulated tension/compression histories
31-32	Tension/compression thresholds
33-34	Tension/compression kappaOne
35-36	Tension/compression kappaTwo
37-38	Non-viscous tensile/compressive damage
39	Selected characteristic length, mm
40	Frame identifier 386
41-49	Accumulated rotation matrix, Fortran column-major
50	Accumulated trapezoidal stress work: dot((S_old+S_new)/2,DSTRAN)
51	Diagnostic dot(S_new,total_strain-viscous_plastic)/2
52	SDV50-SDV51, not validated plastic/damage dissipation
53-58	Viscous plastic strain, engineering shear
59-60	Viscous tensile/compressive damage
61	Viscous schema identifier 1049
62	mu; saved/input consistency checked
63–65	Geometry schema 2, node count (6 or 8), reserved zero
66–89	Initial corner-node coordinates, global reference frame; unused wedge coordinates are zero

11.3 Numerical acceptance controls

The accepted damage increase per increment is limited to 0.25: non-viscous damage for mu=0, or the actual relaxed damage entering stress when mu>0. This is an increment-control limit, not a final damage cap. The normalized difference between two converged Newton solutions is limited to 0.001. Rejection preserves stress and STATEV and requests PNEWDT=0.5. Stress and hardening-normalized yield residual tolerances remain 1e-12. The kappa residual uses the larger of 1e-12, 1e-8 times the absolute kappa increment, and eight floating-point spacings at the current/old kappa magnitude. Thus large accumulated kappa does not impose an unattainable absolute precision. Tensile damage STATEV37/59 displays a damage distribution, not discrete crack directions. Do not use UMAT PE/PEEQ as substitutes for implemented plastic state variables.

12 Application limits and distribution

12.1 Tangent and energy limitations

For zero-strain-increment queries, an internal-variable-fixed approximate tangent has known differences from full differences including time relaxation. The characteristic length is fixed during differentiation. A known zero-increment query has relative difference 0.436 from full numerical differences. Full tangent consistency across every branch is not claimed. SSE denotes elastic strain energy, SPD plastic dissipation and SCD creep dissipation[8]. This UMAT does not update these standard energy outputs. They cannot be used to evaluate an energy balance or energy consumed by fracture. STATEV50–52 are diagnostics, not a validated dissipation decomposition. Missing energy outputs do not mean the stress update is absent.

12.2 Contribution and distribution

Share small reproducible examples, environment, material values/units, loading history, viscosity and logs; omit confidential models and license credentials. Source, added tools and explanations are LGPL-2.1-or-later, without warranty. Preserve LICENSE and THIRD_PARTY_NOTICES.md on redistribution. No OOFEM or SIMULIA endorsement is implied. Third-party papers and experimental plots retain their owners' rights. Comparison plots are independently generated from simulation results and numerical values read from published plots; original third-party images and measured datasets are not distributed.

References

[1] P. Grassl, D. Xenos, U. Nyström, R. Rempling, K. Gylltoft (2013). CDPM2: A damage-plasticity approach to modelling the failure of concrete. International Journal of Solids and Structures 50, 3805–3816. https://arxiv.org/abs/1307.6998. Figures 11 and 12: PDF p.22. The corrigendum is also considered.

[2] P. Grassl, M. Jirásek (2006). Damage-plastic model for concrete failure. International Journal of Solids and Structures 43, 7166–7196.

[3] OOFEM, concretedpm2.C and concretedpm2.h. Reference commit `1c1dbd2b4b8c95dcc26878bf5009d44ed535e1a6`. https://github.com/oofem/oofem.

[4] I. Imran, S.J. Pantazopoulou (1996). Experimental study of plain concrete under triaxial stress. ACI Materials Journal 93, 589–601. Used through Grassl et al., Fig.11, not original numeric data. The corrigendum cites Imran (1994); correspondence remains unresolved.

[5] F.C. Caner, Z.P. Bažant (2000). Microplane model M4 for concrete: II. Algorithm and calibration. Journal of Engineering Mechanics 126, 954–961. Experimental reference cited in Grassl et al., Fig.12; readings here are from that reproduced figure.

[6] P.E. Petersson (1981). Crack Growth and Development of Fracture Zones in Plain Concrete and Similar Materials. Report TVBM-1006, Lund University. https://lup.lub.lu.se/record/1272935.

[7] SIMULIA Abaqus 2026, Benchmarks, Notched unreinforced concrete beam under 3-point bending, Fig.11. `SIMACAEBMKRefMap/simabmk-c-notchedconcbeam.htm`. Direct reading source for the Petersson markers.

[8] SIMULIA Abaqus 2026, User Subroutines Reference: UMAT (STATEV, SSE, SPD, SCD; stress/strain components; convergence rate), `SIMACAESUBRefMap/simasub-c-umat.htm`; UEXTERNALDB (LOP=4), `SIMACAESUBRefMap/simasub-c-uexternaldb.htm`. Keywords: CONTACT CONTROLS, STIFFNESS SCALE FACTOR, `SIMACAEKEYRefMap/simakey-r-contactcontrols.htm`.

[9] SIMULIA Abaqus 2026, Materials, Concrete damaged plasticity. `SIMACAEMATRefMap/simamat-c-concretedamaged.htm`.